



Trail Making Test: Comparison of paper-and-pencil and electronic versions

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ABSTRACT

The Trail Making Test (TMT) was adapted for the iPad by Parker-O'Brien, which uses the 2004 Tombaugh norms. This study investigated the equivalency of this electronic test by (a) examining the test-retest reliability of the iPad-TMT, and (b) calculating the concurrent validity between the two versions. The sample included 77 healthy adults. Reliability was assessed by Pearson product-moment correlation and intraclass correlation coefficient, while validity was assessed by MANOVA. Results indicate that Part A of the iPad-TMT did not demonstrate adequate test-retest reliability over 1 week ($r = 0.15\text{--}0.70$); Part B demonstrated adequate test-retest reliability in the majority of groups ($r = 0.33\text{--}0.80$). Conversely, Part A of the electronic TMT demonstrated adequate concurrent validity, whereas Part B did not; however, validity in Part A has minimal significance without adequate reliability. Handedness had a significant effect on performance, with left-handers performing slower on the electronic TMT Part A ($p < .05$) and the traditional TMT Part B ($p < .05$). Clinicians should use caution when using electronic versions of traditional tests, as they may assess different constructs. New norms should be developed. The role of handedness on TMT performance should be further assessed.

KEYWORDS

Computerized applications;
neuropsychological tests;
neuropsychology; test
construction

Introduction

The Trail Making Test (TMT) is a commonly used neuropsychological instrument (Camara, Nathan, & Puente, 2000; Rabin, Barr, & Burton, 2005), released in several different versions, including paper-and-pencil and electronic forms. This includes a recently developed electronic version of the TMT (iPad-TMT; Parker-O'Brien and Associates, 2013a, 2013b), which uses norms intended for the paper-and-pencil version (PAP-TMT; Tombaugh, 2004), and has yet to be verified against the well-accepted Halstead Reitan version of the TMT (Reitan & Wolfson, 1993).

The PAP-TMT (Reitan & Wolfson, 1993) is a neuropsychological measure that requires examinees to connect numbered and lettered circles. The first condition requires examinees to connect numbers in sequential order from one to 25, as quickly as possible. The second condition is similar but examinees are asked to alternate between matching numbers and letters in sequential order, with numbers from one to 13 and letters A through L. The total time to completion is converted into a standardized score, with several

normative sets available (see Mitrushina, Boone, Razani, & D'Elia, 2005, for a review). While the number of errors is not converted to a standardized score, errors contribute to the overall seconds to completion as participants are directed to correct the mistake while completing the task. The TMT measures a number of different cognitive constructs, including divided attention (Lezak, Howieson, Bigler, & Tranel, 2012), scanning (Lyneess, Eaton, & Schneider, 1994; Stuss, Stethem, & Poirier, 1987), visuomotor tracking (Schear & Sato, 1989; Shum, McFarland, & Bain, 1990), cognitive flexibility (Greenlief, Margolis, & Erker, 1985; La Rue, 1992), working memory (Crowe, 1998), and speed of mental processes (Strauss, Sherman, & Spreen, 2006). Performance on the PAP-TMT has been influenced by age and education (Mitrushina et al., 2005), IQ (Diaz-Asper, Schretlen, & Pearlson, 2004), and cultural factors (Horton & Roberts, 2003; Lee, Cheung, Chan, & Chan, 2000; Lucas et al., 2005; Manley et al., 1998; Roberts & Horton, 2001).

Computer-based technologies are increasingly more common in neuropsychology (Parsey & Schmitter-Edgecombe, 2013), with benefits of strict

standardization of administration, immediate scoring, accurate time recording, and secure data storage (Cernich, Brennana, Barker, & Bleiberg, 2007). With each assessment conversion, however, there remain concerns regarding equivalency to the traditional counterpart (Bauer et al., 2012; Parsey & Schmitter-Edgecombe, 2013). Computerized assessments can produce significant changes in response, measure different cognitive demands than their traditional counterparts, and are considered by some to be a new generation of tests (Bauer et al., 2012; Crook, Kay, & Larrabee, 2009).

To date, several neuropsychological tests have been converted to computerized formats, but with variable psychometric results. A collection of neurobehavioral tests known as the Behavioral Assessment and Research System (BARS; Anger, Rohlman, & Sizemore, 1994; Anger et al., 1996; Rohlman, Sizemore, Anger, & Kovera, 1996) has shown variable test-retest reliability, from low to high, over the course of 6 hours to 1 week (Campbell et al., 1999; Farahat, Rohlman, Storzbach, Ammerman, & Anger, 2003). Similarly, a computerized version of the Continuous Visual Memory Test (CVMT; Trahan & Larrabee, 1988) showed variable reliability, from low to adequate. Although low reliability could be due to sample characteristics (Baños, Dickson, & Greer, 2001), it is also likely that computerized tests may require longer sampling of behavior to assess the desired constructs.

While few studies have assessed the test-retest reliability of converted measures, many studies examined the validity of these measures by comparing them to their traditional versions. Several tests demonstrated similar performance between mediums, including the mental rotation task (Voyer et al., 2006), CVMT (Baños et al., 2001; Trahan & Larrabee, 1988), Mill Hill vocabulary test (French & Beaumont, 1990; Raven, Court, & Raven, 1998), Tower of Hanoi (Mataix-Cols & Bartré-Faz, 2002; Simon, 1975), Raven's progressive matrices in a nonclinical sample (Raven, 1958; Williams & McCord, 2006), Test of Memory Malinger (Tombaugh, 1996; Vanderslice-Barr, Miele, Jardin, & McCaffrey, 2011), and two subtests from the BARS (Campbell et al., 1999). The Halstead Category Test (Reitan & Wolfson, 1993) demonstrated adequate concurrent validity in some studies (Choca & Morris, 1992; Mercer, Harrell, Miller, Childs, & Rockers, 1997), but not in with clinical populations (Berger, Chibnall, & Gfeller, 1997).

A number of other neuropsychological measures which have been converted to computer were able to establish concurrent validity for some, but not all,

obtained scores, including line orientation (Aşkar et al., 2012; Benton, 1978), Raven's progressive matrices (Raven, 1958) in a clinical sample (French & Beaumont, 1990), Wisconsin Card Sorting Test (Feldstein et al., 1999; Fortuny & Heaton, 1996; Heaton, Chelune, Talley, Kay, & Curtiss, 1993; Steinmetz, Brunner, Loarer, & Housseman, 2010; Tien et al., 1996), BARS (Campbell et al., 1999), Word Memory Test (Green, 2005; Hoskins, Binder, Chaytor, Williamson, & Drane, 2010), Enhanced Cued Recall Test (Aşkar et al., 2012; Solomon et al., 1998), and Benton Visual Retention Test (Sivan, 1991; Thompson, Ennis, Coffin, & Farman, 2007). Interestingly, these studies have described additional relevant considerations regarding these converted measures, including change in performance when being observed (Yantz & McCaffrey, 2007), increased administration time with tablet version (Snowdon, Hussein, Kent, Pino, & Hachinski, 2014), participants responding more quickly to electronic formats (Mataix-Cols & Bartré-Faz, 2002), and preference for a specific format of the test (Thompson et al., 2007).

Specific to the TMT, several studies have attempted to employ electronic formats, though with variable results. Salthouse and Fristoe (1995) used a computerized version of the TMT in which participants used the arrow keys to direct the cursor on the screen to move from circle-to-circle. While estimated reliability of the computerized-TMT was high and the correlation between the two versions was also high, the authors did not utilize counterbalancing, nor address the impact of practice effects on performance. Additionally, the stimulus and various conditions were substantially different from the PAP-TMT. An fMRI study (Zakzanis, Mraz, & Graham, 2005) which utilized a tablet-version of the TMT did not assess reliability or validity in the measure and the task demands and visual array were substantially different from the PAP-TMT. Two separate studies (Dahmen, Cook, Fellows, & Schmitter-Edgecombe, 2017; Fellows, Dahmen, Cook, & Schmitter-Edgecombe, 2017) utilized a tablet based version of the TMT which demonstrated acceptable levels of correlation between the tablet version and the PAP-TMT; however, neither of these studies assessed the reliability of the table version prior to assessing validity. Without understanding the reliability of the measure, these validity findings are largely meaningless (Sherman, Brooks, Iverson, Slick, & Strauss, 2011), although the authors do offer this as a limitation. Other limitations included fewer circles on the tablet, no counterbalancing between versions, and the effects of practice on performance was not addressed.

Some argue that the advantages of computerized administration outweigh the need to establish psychometric equivalency (Hoskins et al., 2010). In line with this argument, some of the computerized versions of popular neuropsychological tests are widely used (Feldstein et al., 1999). The most notable example is perhaps the computerized WCST (Heaton & PAR Staff, 2003). The computerized version (WCST: CV4; Heaton & PAR Staff, 2003) is scored using the norms for the traditional version (Heaton et al., 1993). Many studies comparing the two versions failed to establish concurrent validity for all the test scores. Fortuny and Heaton (1996) found significant differences in the amount of time to complete the first category, with the computerized version taking longer. Feldstein et al. (1999) compared the traditional WCST (Heaton et al., 1993) to four different computer conditions and found that none of the four versions produced adequate concurrent validity on all scores. Steinmetz et al. (2010) found that the variance of percentage of errors, perseverative errors, and failure to maintain set differed substantially between versions. Tien et al. (1996) found marginally significant differences in perseverative errors and significant differences in perseverative responses, total errors, total correct, and nonperseverative errors. Feldstein et al. (1999) found significant differences between the touch-screen and traditional versions of the WCST (Heaton et al., 1993) with much greater differences on this type of touch-response than with a conventional mouse. Despite these studies, computerized WCST (Heaton et al., 1993) remains popular among clinicians, suggesting that neuropsychologists are eager to benefit from the advantages of computerized administration, even in the face of psychometric shortcomings.

The PAP-TMT (Reitan & Wolfson, 1993) includes two separate conditions. The first condition requires examinees to connect numbers in sequential order, as quickly as possible. In the second condition, examinees are asked to alternate between matching numbers and letters in sequential order. There are several available normative data sets to assist with scoring, with Tombaugh (2004) scores being one of the most comprehensive. Parker-O'Brien and Associates developed an iPad based application of the TMT in 2013 (Parker-O'Brien and Associates, 2013a, 2013b) using norms developed by Tombaugh (2004) for the traditional Halstead-Reitan version of PAP-TMT. While these two versions have similar instructions, presentations, and normative analyses, there have been no studies to date that examine the concurrent validity of these two measures. Furthermore, the iPad-TMT

(Parker-O'Brien and Associates, 2013a, 2013b) does not currently have reliability studies.

The present study sought to (1) assess the test-retest reliability of the iPad-TMT (Parker-O'Brien and Associates, 2013a, 2013b) in healthy adults over 1 week and (2) compare the traditional Halstead-Reitan version of the PAP-TMT (Reitan & Wolfson, 1993) to the iPad-TMT (Parker-O'Brien and Associates, 2013a, 2013b) to verify if there are differences in performance when tested in the same population. Considering current knowledge, it is hypothesized that when testing a normal population, the PAP-TMT (Reitan & Wolfson, 1993) and the iPad-TMT (Parker-O'Brien and Associates, 2013a, 2013b) will be significantly different. Additionally, it is hypothesized that the iPad-TMT (Parker-O'Brien and Associates, 2013a, 2013b) will produce low reliability results over 1 week.

Methods

The study was approved by the Institutional Review Board at a Midwest graduate psychology school (Protocol # IRB-15-01-0029 and IRB-14-11-0003). Inclusion criteria included, (a) currently enrolled student and (b) between the ages of 18- and 89-years-old. The study included 117 participants. Z-scores were calculated within each group for each dependent variable, and cases with z-scores greater than 2.5 or less than -2.5 were eliminated from all analyses (Field, 2013). A z-score of 2.5 means that the participant's scores were two and a half standard deviations above the mean, suggesting either very superior or very low performance (Field, 2013). These individuals are assumed to be nonrepresentative of the sample and eliminated from analyses. In order to maintain rigorous statistical standards, participants who obtained any z-score of 2.5 or above were completely eliminated from analyses, as opposed to eliminating only the score that was greater than 2.5 and keeping all other data for that participant. This left 77 participants who provided enough data to be included in the final data analyses.

To understand the possible practice effects, the participants were split into four groups, counterbalanced on order of test completion. Participants were recruited through flyers, announcements, and campus-wide emails. Table 1 displays the groups and the order of administration. Participants completed an informed consent and screening measure to determine eligibility and familiarity with the TMT at the beginning of the first session. Group one ($n=17$) and group two ($n=17$) completed the iPad-TMT only during session one, then returned in 7-days to

Table 1. Participant groups based on order of administration.

Group	Session 1		Session 2	
	1st Assessment	2nd Assessment	1st Assessment	2nd Assessment
1. ($n = 17$)	iPad	–	Paper-and-pencil	iPad
2. ($n = 17$)	iPad	–	iPad	Paper-and-pencil
3. ($n = 22$)	Paper-and-pencil	iPad	iPad	–
4. ($n = 21$)	iPad	Paper-and-pencil	iPad	–

Note. – indicates only one assessment was administered. Time between sessions was 7-days for all groups.

complete the PAP-TMT and the iPad-TMT again. Group three ($n = 22$) and group four ($n = 21$) completed the PAP-TMT and the iPad-TMT in session one, then returned in 7-days to complete the second administration of the iPad-TMT. Part A of the TMT was always administered before Part B, for both the iPad and paper conditions. Participants in each group completed a demographics form at the end of session one. Each session took approximately 10 min.

Instruments

The Halstead-Reitan TMT (Reitan & Wolfson, 1993) was administered utilizing administration instructions from Parker-O'Brien and Associates TMT application (Parker-O'Brien and Associates, 2013a, 2013b) was administered on an iPad using the instructions and demonstration provided with the application. Errors were immediately corrected on both versions by the examiner saying “no,” marking out the incorrect line, and reorienting the participant back to the last correct item. All iPad-TMTs were administered on Apple iPad-Air (9.7-inch diagonal LED-backlit Multi-Touch display with IPS technology) with a resolution of 2048 x 1536 and iOS version 8.4, utilizing the auto-brightness setting. Only one participant asked that the brightness be increased. All participants used a stylus to complete the iPad-TMT. The time to completion for each measure (i.e. PAP-TMT Part A, PAP-TMT Part B, iPad-TMT Part A, iPad-TMT Part B) in seconds was used as the dependent variable, while group was the independent variable. Z-scores calculated by group for each dependent variable were used to eliminate outliers. The effects of demographics on performance were assessed using independent samples *t*-test and a one-way analysis of variance (ANOVA) was conducted to assess the effect of handedness on performance. Test-retest reliability was assessed using Pearson's product-moment correlations (*r*) and intraclass correlation coefficients (ICC). Concurrent validity was assessed using k-group or one-way multivariate analysis of variance (MANOVA).

Table 2. Demographic variables.

Demographics	%
Education (highest degree completed)	
Bachelors	64.0
Masters	36.0
Sex	
Female	75.6
Male	24.4
Ethnicity	
Caucasian	67.4
African American	7.0
Hispanic	2.3
Other*	23.3
Handedness	
Right	83.7
Left	12.8
Ambidextrous	3.5

Note. * identifies self-reported ethnic groups that were represented by less than 2% of the participants and include the following: African American/Native American, Afro-Cuban, Asian, Asian/American, Asian/Chinese, Asian/Hawaiian, Asian/Japanese/Chinese, Caucasian/Asian, Caucasian/Jewish, Caucasian/Latina, Caucasian/Middle Eastern, Caucasian/Native American, Chinese, East Indian, Euro-American, Hispanic, Hispanic/Caucasian, Hispanic/Latina, Hispanic/Latino, Latina, Lebanese, Mexican, Meztisa, Mixed Races/Hispanic/Caucasian, Mixed/Black/White, African American/Caucasian/Persian, Nigerian-American, Polish/Scottish, Southeast Asian, and White/Hispanic.

Results

Demographic information is presented in Table 2. Mean age was 26.94 years ($SD = 4.8$, range 22–49), with 100% of participants completing a Bachelor's degree or higher. The sample was 75.5% female, with 83.7% right-handed, 12.8% left-handed, and 3.5% ambidextrous.

Performance on all dependent variables was not affected by the following variables: gender (all $ps > 0.32$), education (all $ps > 0.14$), history of learning disability (all $ps > 0.08$), history of neurologic condition (all $ps > 0.21$), and history of psychiatric condition (all $ps > 0.06$). Using the liberal Least Significant Differences test, age was not different across all pairwise group comparisons (all $ps > 0.32$). Additionally, age was not significantly correlated with performance on any of the tasks, including in the entire sample (all $ps > 0.20$) and within each of the four groups (all $ps > 0.10$). Participants who indicated on the screening questionnaire that they had previously practiced or administered the TMT performed significantly faster

Table 3. Part A test–retest reliability iPad-TMT.

Group	Time 1 ^a	Time 2 ^a	<i>r</i>	ICC	95% CI
1. (<i>n</i> = 17)	24.1 ± 4.0	20.2 ± 3.4	0.157	0.155	[−0.338, 0.581]
2. (<i>n</i> = 17)	23.4 ± 4.2	19.6 ± 3.7	0.500	0.495	[0.035, 0.782]
3. (<i>n</i> = 22)	22.6 ± 4.7	19.3 ± 4.7	0.575	0.575	[0.264, 0.778]
4. (<i>n</i> = 21)	23.3 ± 6.3	18.8 ± 4.6	0.709	0.676	[0.382, 0.845]

Note. *r*: Pearson product moment correlation; ICC: intraclass correlation coefficient; CI: confidence interval. ^aValues shown as mean ± standard deviation.

on PAP-TMT Part A ($M = 17.76$ seconds, $SD = 3.54$) than those that did not ($M = 22.20$ seconds, $SD = 6.6$), $t(64) = -3.92$, $p < .05$. There was a significant effect of handedness on iPad-TMT Part A at time one, $F(2, 83) = 5.85$, $p < .05$, with both right-handed ($M = 22.99$ seconds, $SD = 4.72$, $p < .05$) and ambidextrous ($M = 17.00$ seconds, $SD = 1.00$, $p < .01$) individuals performing faster than left-handed individuals ($M = 26.73$, $SD = 4.78$). There was also a significant effect of handedness on PAP-TMT Part B, $F(2, 6.99) = 12.79$, $p < .05$, with ambidextrous individuals ($M = 32.00$, $SD = 3.61$) performing significantly faster than left-handed individuals ($M = 53.09$, $SD = 21.07$, $p < .03$).

Test-retest reliability analyses

The Pearson product-moment correlations (r) and relative consistency using intraclass correlation coefficients (ICC) for Part A of the iPad-TMT are presented in Table 3. While many authors have suggested different ranges for reliability values, the following results were evaluated using Kline's (1999) criteria, with values less than 0.70 as "low," values between 0.70 and 0.89 as "adequate," and values of 0.90 or above as "high." Pearson r correlations between time 1 and time 2 ranged from .16 to .71. Only group four was able to demonstrate adequate [$r(22) = 0.71$] test–retest reliability for the iPad-TMT over 1-week. Intraclass correlation coefficients reflect lower reliability than Pearson r across all groups. In fact, not one group was able to reach the level of "adequate" test–retest reliability, with values ranging from 0.16 to 0.68. Group two represents the most conventional form of test–retest, as there was no interference from other conditions between time 1 and time 2, resulting in lower effects of practice; however, even in this condition, Pearson $r(15) = 0.500$ and ICC = 0.495. For details, refer to Table 3.

Given that participants who had previously practiced or administered the TMT performed faster on the PAP-TMT, further analyses were run with these individuals excluded, which removed four participants from groups one ($n = 13$), two ($n = 13$), and four

Table 4. Part B test–retest reliability iPad-TMT.

Group	Time 1 ^a	Time 2 ^a	<i>r</i>	ICC	95% CI
1. (<i>n</i> = 17)	53.9 ± 15.2	42.5 ± 12.4	0.326	0.319	[−0.175, 0.685]
2. (<i>n</i> = 17)	50.5 ± 10.1	42.9 ± 9.0	0.800	0.794	[0.519, 0.920]
3. (<i>n</i> = 22)	36.5 ± 9.5	34.4 ± 10.5	0.695	0.692	[0.436, 0.845]
4. (<i>n</i> = 21)	46.1 ± 11.7	35.8 ± 10.2	0.736	0.729	[0.468, 0.873]

Note. *r*: Pearson product moment correlation; ICC: intraclass correlation coefficient; CI: confidence interval. ^aValues shown as mean ± standard deviation.

($n = 21$), and no participants from group 3 ($n = 22$). Without these individuals, group one ($r(11) = 0.20$) and four ($r(20) = 0.82$) demonstrated higher Pearson r correlations; however, again only group four demonstrated acceptable levels of test–retest reliability.

The correlations and ICCs for Part B of the iPad-TMT are presented in Table 4. Pearson r correlations for three of the four groups produced values above acceptable limits – group one [$r(15) = 0.33$], group two [$r(15) = 0.80$], group three [$r(26) = 0.70$], and group four [$r(22) = 0.74$]. Again, group two—the most direct measure of test–retest reliability—revealed the highest correlation for Part B of iPad-TMT. ICCs reflected lower reliability than Pearson's r and only two groups were able to show acceptable levels of ICCs over 1-week: group two (ICC = 0.79) and group four (ICC = 0.73).

Concurrent validity analyses

A one-way between-subjects multivariate analysis of variance (MANOVA) was conducted on three dependent variables: iPad-TMT Part A at time 1, iPad-TMT Part A at time 2, and PAP-TMT Part A. The independent variable was group. A significant p -value (e.g., $p < .05$) is suggestive of a difference in obtained scores, which would suggest the two versions are not equivalent. The assumption of multivariate normality was met by removing outliers that were 2.5 standard deviations above the mean and removal of outliers by group on each dependent variable. Box's M was not significant ($p = .059$), meeting the assumption of homogeneity of covariance matrices. Using Wilks's criterion the composite dependent variate was significantly affected by group (Wilks's λ , $F(9, 71) = 2.18$, $p < .05$, $\eta_p^2 = 0.083$). Univariate ANOVAs were conducted on each dependent measure separately to determine the locus of the statistically significant multivariate effect. These analyses indicated that group significantly affected performance on iPad-TMT Part A at time 2, $F(3, 73) = 3.34$, $p < .05$, partial $\eta^2 = .121$. Perhaps more meaningful is lack of significance between group and both iPad-TMT Part A at time 1, $F(3, 73) = 0.198$, $p > .05$, partial $\eta^2 = 0.061$,

and PAP-TMT Part A, $F(3, 73) = 0.88$, $p > .05$, partial $\eta^2 = 0.035$. LSD post hoc tests suggested that group one ($M = 20.2$, $SD = 3.5$) performed significantly slower on iPad-TMT Part A at time 2 than group three ($M = 17.7$, $SD = 2.5$; $SE = 1.02$, $p < .05$) and four ($M = 17.6$, $SD = 3.0$; $SE = 1.03$, $p < .05$). This difference could be caused by interference from taking the PAP-TMT just before the iPad-TMT at time 2 for group one. The other groups, which showed fairly similar scores, took the iPad-TMT at time 2 without any practice for the past 7 days. There were no differences between other groups (all $ps > 0.05$).

A separate one-way between-subjects MANOVA was conducted on three dependent variables: iPad-TMT Part B at time 1, iPad-TMT Part B at time 2, and PAP-TMT Part B. The independent variable was group. Again, a significant p -value (e.g., $p < .05$) is suggestive of a difference in obtained scores, which would suggest the two versions are not equivalent. The assumption of multivariate normality was met by removing outliers that were 2.5 standard deviations above the mean and removal of outliers by group on each dependent variable. A statistically significant Box's M test ($p < .001$) indicated unequal variance-covariance matrices of the dependent variables across levels of group and thus necessitated the use of Pillai's trace in assessing the multivariate effect and greater reliance on univariate tests of between-subjects effects. Using Pillai's trace, the dependent variate was significantly affected by group, Pillai's trace = 0.627, $F(9, 219) = 6.43$, $p < .001$, partial $\eta^2 = 0.21$. Univariate ANOVAs were conducted on each dependent measure separately to determine the locus of the statistically significant multivariate effect. Group significantly affected performance on all three measures – iPad-TMT Part B at time 1, $F(3, 73) = 14.15$, $p < .001$, partial $\eta^2 = 0.37$; iPad-TMT Part B at time 2, $F(3, 73) = 9.44$, $p < .001$, partial $\eta^2 = 0.28$; PAP-TMT Part B, $F(3, 73) = 4.28$, $p < .05$, partial $\eta^2 = 0.15$. Tamhane post hoc tests were used because these are the most appropriate for when heterogeneity of variance is present. The most prominent differences were between group three on iPad-TMT Part B at time 1 and all other groups. Participants in group three consistently performed faster ($M = 32.8$, $SD = 6.4$) than group one ($M = 53.9$, $SD = 15.2$; $SE = 3.55$, $p < .001$), group two ($M = 50.5$, $SD = 10.1$; $SE = 3.55$, $p < .001$), and group four ($M = 44.5$, $SD = 11.5$; $SE = 3.36$, $p = .001$). Group three had the unique advantage of being the only group that completed the PAP-TMT just before the first administration of the iPad-TMT. Similarly, there were significant differences between group on the PAP-TMT

Part B. Participants in group four ($M = 37.4$, $SD = 6.2$) performed faster than group 1 ($M = 46.9$, $SD = 7.6$; $SE = 2.73$, $p = .001$) and group 3 ($M = 43.3$, $SD = 10.4$; $SE = 2.55$, $p < .05$). Group four also had the unique advantage of completing the iPad-TMT just before the PAP-TMT; however, this advantage is not as obvious as in the effect on the iPad-TMT, as not all groups showed the benefit of this practice. While there were not as many significant differences with group and iPad-TMT Part B at time 2, group three ($M = 30.5$, $SD = 5.2$) did perform significantly faster than group two ($M = 42.9$, $SD = 9.0$; $SE = 2.87$, $p < .001$). There were no differences between other groups (all $ps > 0.18$).

Discussion

The study examined the test-retest reliability and concurrent validity of an electronic version of the TMT. Overall, the results suggest questionable reliability of the Parker-O'Brien TMT (2013), and insufficient concurrent validity when contrasted with the PAP-TMT.

Reliability

Part B of the iPad-TMT demonstrated stronger test-retest reliabilities, with three of the four groups demonstrating adequate values. These findings are consistent with previous research showing that Part B demonstrates more stable test-retest reliabilities than Part A (Bornstein, Baker, & Douglas, 1987; Dikmen, Heaton, Grant, & Temkin, 1999; Levine, Miller, Becker, Selnes, & Cohen, 2004; Matarazzo, Wiens, Matarazzo, & Goldstein, 1974; Mitrushina & Satz, 1991). Conversely, iPad-TMT Part A showed poor test-retest reliability. Specifically, one out of four groups was able to demonstrate adequate test-retest reliability with Pearson's r over 1 week for Part A. When assessing test-retest reliability with ICCs, no group was able to obtain adequate reliability on Part A, regardless of the participants' familiarity with the test.

The implications are trifold; first, the construct or medium of the electronic TMT Part A lacks reliability and should be redesigned to improve its temporal stability. Second, it appears that iPad-TMT Part B has stronger reliability and may, if these results are replicated, be used in electronic forms. Finally, when considering reliability of the test, the iPad-TMT is not recommended for clinical use in its current form.

Validity

Since the iPad-TMT showed adequate test–retest reliability for some of the groups, the concurrent validity data was analyzed. Part A demonstrated concurrent validity when comparing iPad-TMT at time 1 to PAP-TMT. Of course, since iPad-TMT Part A demonstrated poor reliability, these findings are largely meaningless (Sherman, Brooks, Iverson, Slick, & Strauss, 2011). Comparison of iPad-TMT scores at time 2 with PAP-TMT showed significant differences in performance. In addition, Part B lacked concurrent validity for all groups.

Poor concurrent validity, even for aspects of the TMT with adequate reliability, is not surprising. The iPad-TMT and PAP-TMT, despite their similarities, are quite different. The major difference appears to be how individuals interact with each medium. Participants in the current study completed the iPad-TMT with a stylus, in an attempt to closely mimic the PAP-TMT. Many participants commented that they do not regularly use a stylus when working with a tablet and thus may have felt uncomfortable with this method. While the array of numbers and letters on the iPad-TMT is fairly close to the PAP-TMT, the difference of nearly five additional inches on the diagonal of the paper compared to the iPad results in changes in proximity and arrangement of stimuli between the two versions of the TMT. The change in the spatial array may even be the cause of the differences in performance due to implicating different cognitive and/or motor mechanisms. Previous studies have found that the format of the test can have a significant impact on performance, such as increased administration time with a tablet (Snowdon, Hussein, Kent, Pino, & Hachinski, 2014), quicker responses to electronic formats (Mataix-Cols & Bartré-Faz, 2002), significant differences when drawing on a tablet (Thompson et al., 2007), and significant differences between touch screen and computer mouse on the computerized WCST (Feldstein et al., 1999; Heaton et al., 1993). Interacting with paper-pencil/physical test stimuli is likely a different kind of human experience than interacting with a computer/touch screen. Given these results, it is strongly advised that the iPad-TMT and PAP-TMT not be used interchangeably.

Practice effects

Perhaps most important in the assessment of the TMT reliability and validity is the effect of practice, which impacts both the test–retest reliability and concurrent validity. It is accepted that neuropsychological practice effects impact performance in the first

12 months (Basso, Bornstein, & Lang, 1999), although PAP-TMT appears to be affected regardless of retest intervals (Bornstein et al., 1987; Barr, 2003; Dye, 1979; Stuss et al., 1987; McCaffrey, Duff, & Westervelt, 2000). In our study, group one performed slower than group three and four on iPad-TMT Part A at time 2. This was the only group that was administered the PAP-TMT just before the second administration of the iPad-TMT. A possible explanation is that the PAP-TMT actually interfered with performance on the iPad-TMT. While the spatial arrangement of the numbers on the two versions is fairly close, the dimensions of the iPad-TMT are smaller than the PAP-TMT. It may be that the differences in the spatial arrangement were great enough to negatively affect performance in this group in particular. If the two versions of the test are indeed equivalent, prior studies would lead us to believe that we would see the same practice effects, even across different versions. The fact that we did not observe the same practice effects in group one—in fact their performance actually declined—lends further support to the idea that the two versions are not equivalent.

Handedness

In the present study, the effect of handedness was not only present on the iPad-TMT but the traditional PAP-TMT as well, which was an unexpected finding given the vast amount of research on the traditional TMT. It is perhaps not surprising. The instructions listed in Strauss et al. (2006) instruct the participant to keep their pencil on the page throughout the administration; therefore, participants will frequently block stimuli with their writing hand. Although right- and left-handed individuals block almost the exact same amount of the stimuli during administration for both versions, the next stimulus in sequence is frequently on the left of the writing hand, making it more difficult for the left-handed individuals to notice them. To accommodate for this, left-handers often raise their hand to scan the page below, likely adding to the final time score. Finally, being aware that some of the stimuli may be hidden beneath one's writing hand and trying to adjust for that, may impact the thought process of the individual.

Future directions

Future iPad TMT research should examine other forms of reliability, including internal consistency and perhaps alternate form reliability to decrease practice

effects, in diverse groups of participants. Even if the iPad-TMT is found to be reliable and can also obtain adequate concurrent validity with the PAP-TMT, separate data should be reported for the iPad-TMT rather than using the PAP-TMT norms, in accordance with the American Academy of Clinical Neuropsychology (AACN) and the National Academy of Neuropsychology (NAN) guidelines (Bauer et al., 2012). Future research should also assess the role of handedness on TMT performance, to identify the need for separate handedness norms.

Limitations

There are a number of limitations in the current study. All participants completed all phases of the study, resulting in an increased risk of practice effects. Demographics were not representative of the population of the United States, affecting generalizability. The relatively small effect sizes of group on part A did not reach statistical significance despite accounting for notable amounts of variance in completion time (3.5% for iPad time 1 and 6.1% for PAP time 1). This omnibus effect would require a larger sample to reach statistical significance ($N=172$ and $N=308$, respectively), despite the potential clinical significance of choosing one test over the other. In particular, the difference between the iPad and paper-and-pencil groups at time 1 represents a medium effect size, Hedges's $g=0.495$ (Cohen, 1992; Durlak, 2009), suggesting a meaningful difference between the two versions despite a statistically non-significant difference between these two conditions ($p=.15$). If this effect is indeed real, 51 participants per group would be required to detect it at $\alpha=0.05$ and $\beta=0.2$. Thus, in all, future studies should use larger, representative samples.

Conclusions

Given the variability of results in the present study, we strongly recommend against interchanging the iPad-TMT and PAP-TMT. Normative data that have been developed for the PAP-TMT should not be used in interpretation of the iPad-TMT. While the iPad-TMT compares the Tombaugh (2004) normative data to the current users' performance for scoring purposes, these norms should not be used given these findings. To consider the iPad-TMT as a valid clinical tool, new norms need to be developed to meet acceptable professional standards.

Disclosure statement

The authors report no conflicts of interest.

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